

RIVERINE FLOODS AND URBAN FLOOD RESILIENCE

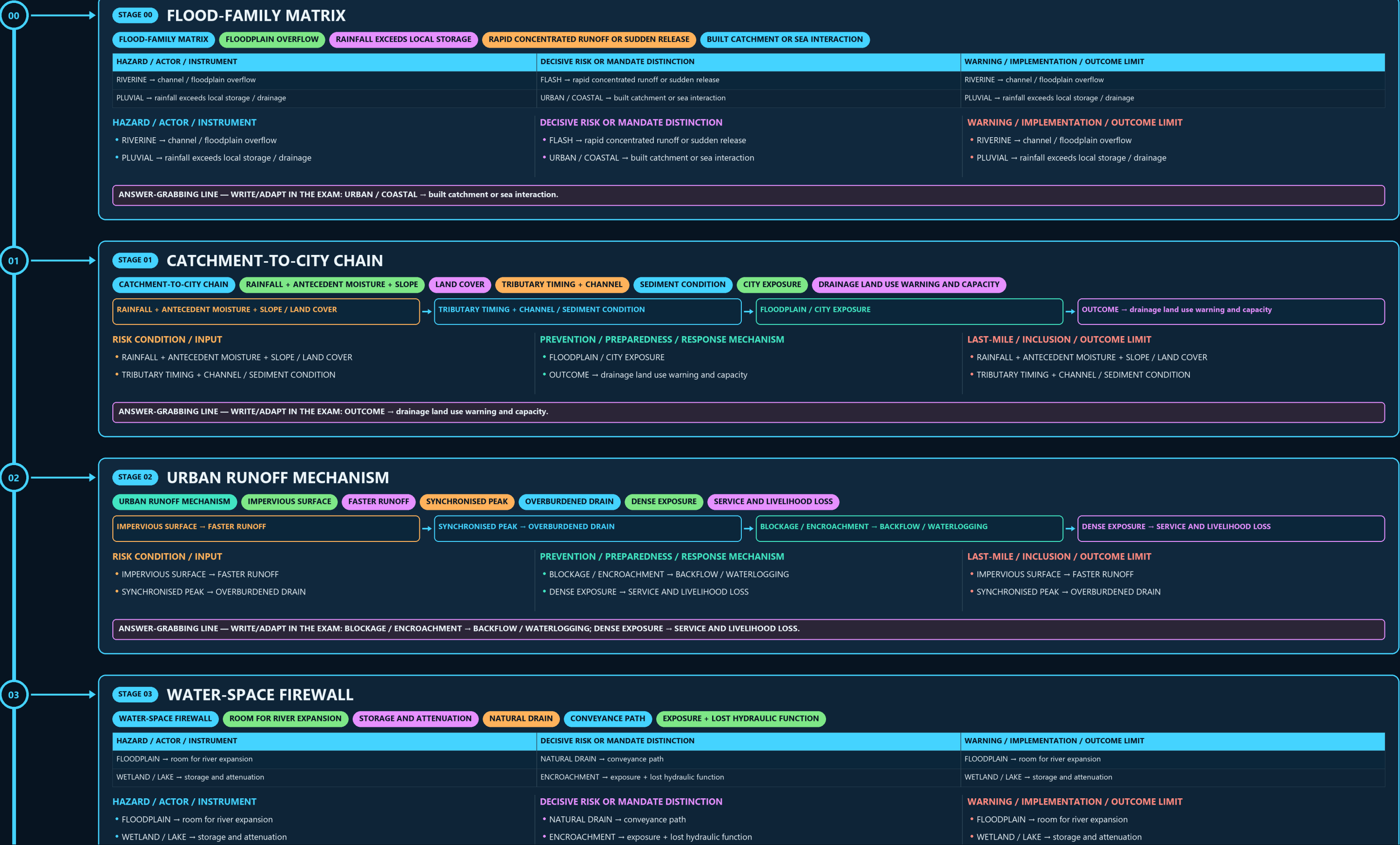
FLOOD-FAMILY MATRIX → CATCHMENT-TO-CITY CHAIN → URBAN RUNOFF MECHANISM → WATER-SPACE FIREWALL → COMPOUND COASTAL FLOOD MAP → CWC FORECAST RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

STAGE 03

WATER-SPACE FIREWALL

WATER-SPACE FIREWALL

ROOM FOR RIVER EXPANSION

STORAGE AND ATTENUATION

NATURAL DRAIN

CONVEYANCE PATH

EXPOSURE + LOST HYDRAULIC FUNCTION

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
FLOODPLAIN → room for river expansion	NATURAL DRAIN → conveyance path	FLOODPLAIN → room for river expansion
WETLAND / LAKE → storage and attenuation	ENCROACHMENT → exposure + lost hydraulic function	WETLAND / LAKE → storage and attenuation

HAZARD / ACTOR / INSTRUMENT

- FLOODPLAIN → room for river expansion
- WETLAND / LAKE → storage and attenuation

DECISIVE RISK OR MANDATE DISTINCTION

- NATURAL DRAIN → conveyance path
- ENCROACHMENT → exposure + lost hydraulic function

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- FLOODPLAIN → room for river expansion
- WETLAND / LAKE → storage and attenuation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NATURAL DRAIN → conveyance path; ENCROACHMENT → exposure + lost hydraulic function.

04

STAGE 04

COMPOUND COASTAL FLOOD MAP

COMPOUND COASTAL FLOOD MAP

RIVER FLOW

PLUVIAL RUNOFF

URBAN DRAIN

DOWNSTREAM IMPEDANCE

COMPOUND INUNDATION

RIVER FLOW → ESTUARY

PLUVIAL RUNOFF → URBAN DRAIN

SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE

COMBINATION → COMPOUND INUNDATION

HAZARD / RISK / RESILIENCE CONCEPT	INSTITUTIONAL OR COMMUNITY RESPONSE	CAPACITY / STATUS / EVIDENCE LIMIT
- RIVER FLOW → ESTUARY - PLUVIAL RUNOFF → URBAN DRAIN	- SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE - COMBINATION → COMPOUND INUNDATION	- RIVER FLOW → ESTUARY - PLUVIAL RUNOFF → URBAN DRAIN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE.

05

STAGE 05

CWC FORECAST RAIL

CWC FORECAST RAIL

1 OBSERVE LEVEL

2 ANALYSE AND FORECAST

3 ISSUE SITE-SPECIFIC CATEGORY

4 ADMINISTRATION

PROJECT AUTHORITY ACTS

1 OBSERVE LEVEL / DISCHARGE / INFLOW

2 ANALYSE AND FORECAST

3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE

4 ADMINISTRATION / PROJECT AUTHORITY ACTS

HAZARD / RISK / RESILIENCE CONCEPT	INSTITUTIONAL OR COMMUNITY RESPONSE	CAPACITY / STATUS / EVIDENCE LIMIT
1 OBSERVE LEVEL / DISCHARGE / INFLOW 2 ANALYSE AND FORECAST	3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE 4 ADMINISTRATION / PROJECT AUTHORITY ACTS	1 OBSERVE LEVEL / DISCHARGE / INFLOW 2 ANALYSE AND FORECAST

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE; 4 ADMINISTRATION / PROJECT AUTHORITY ACTS.

06

STAGE 06

RESERVOIR GOVERNANCE BOUNDARY

RESERVOIR GOVERNANCE BOUNDARY

INFLOW FORECAST + CURRENT STORAGE

OPERATING RULE

DAM CONDITION

DOWNSTREAM CAPACITY

GATE DECISION

TIMELY DOWNSTREAM WARNING

DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE

INFLOW FORECAST + CURRENT STORAGE

OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY

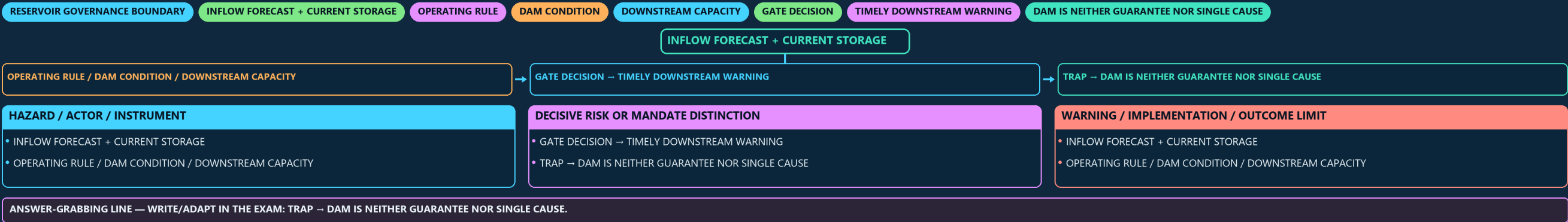
GATE DECISION → TIMELY DOWNSTREAM WARNING

TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE

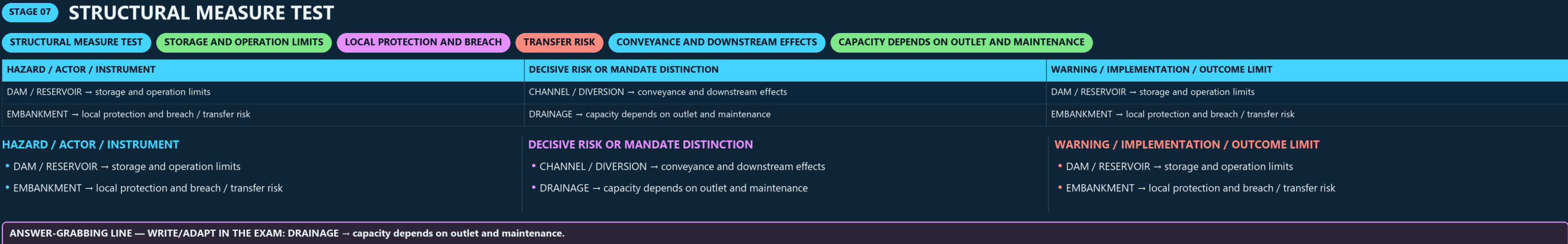
HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
- INFLOW FORECAST + CURRENT STORAGE - OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY	- GATE DECISION → TIMELY DOWNSTREAM WARNING - TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE	- INFLOW FORECAST + CURRENT STORAGE - OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE.

RESERVOIR GOVERNANCE BOUNDARY



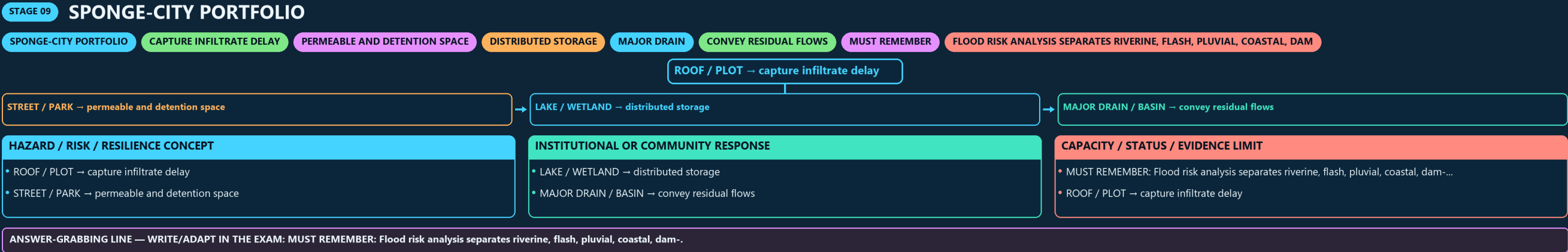
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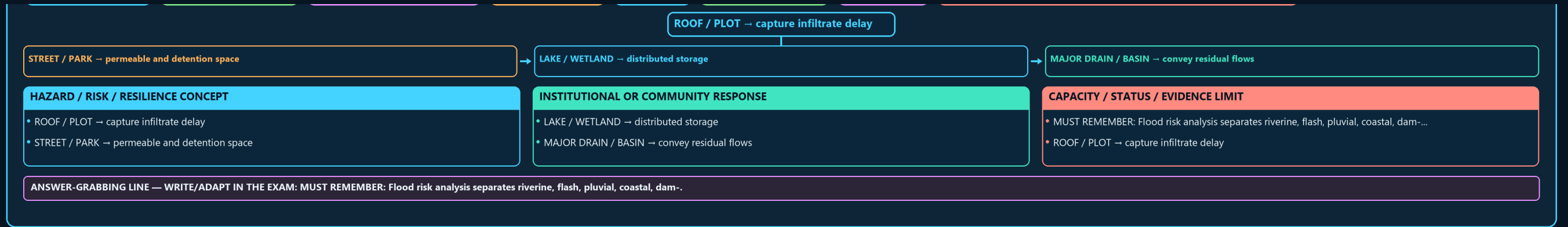


08



09

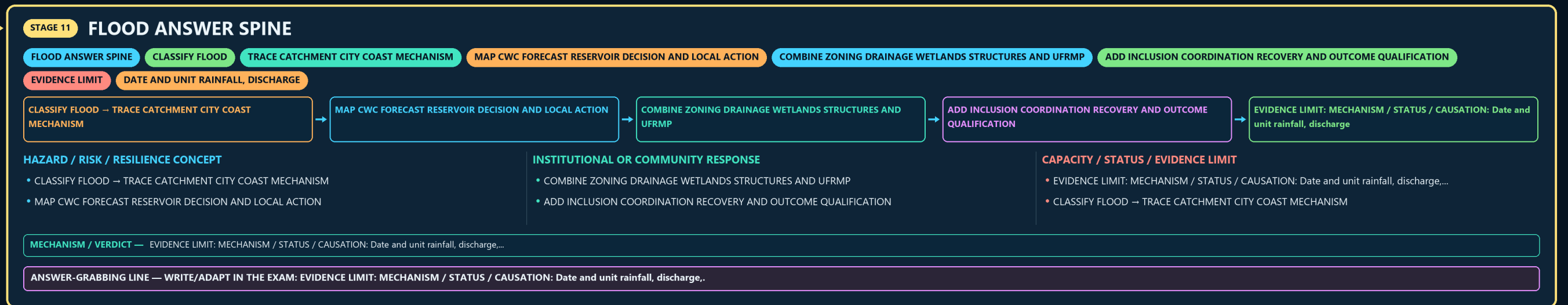




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11



E

